

Problem Set 1: Solutions

Your name

Your Collaborators

Total Number of Points: 40.

Instructions: Solutions should be typeset in $\text{\LaTeX}$ and submitted as PDF by 11:59:59 PM ET on the due date. You can collaborate on the psets in small groups (of up to three), but must write your solutions separately, in your own words. This means not copying from a board nor from your collaborator's computer.

We strongly recommend **not** using AI tools to solve your problem sets. You are responsible for understanding everything you submit and you must be able to explain and justify every part of your solution. In particular, the oral problem set review (10% of the course grade) will evaluate your understanding of the solutions you submit. In all cases, you must list your collaborators, human or machine.

Problem 1. Negligible or not? (8 points) For each of the following functions below, either prove or disprove that it is negligible.

- (a) (2 points) $\mu(n) := 1.01^{-n}$. (**Note:** Here, and throughout the course, all logarithms are in base 2 unless otherwise specified.)

Solution

Negligible. Let p be any polynomial of degree d . For sufficiently large n , $p(n) < n^{d+2}$. Moreover, since an exponential eventually grows faster than any polynomial, for sufficiently large n , $1.01^n > n^{d+2}$. Therefore, for all sufficiently large n ,

$$\mu(n) = \frac{1}{1.01^n} < \frac{1}{n^{d+2}} < \frac{1}{p(n)}.$$

Hence, μ is negligible.

- (b) (2 points) $\mu(n) := e^{-(\log n)^3} + e^{-(\log n)^2} + e^{-\log n}$.

Solution

Not negligible. Indeed, $\mu(n) \geq e^{-\log n}$. Since all logarithms are base 2, $e^{\log n} = n^{\log e} < n^2$, and hence $e^{-\log n} > \frac{1}{n^2}$. Thus $\mu(n) > \frac{1}{n^2}$, so, taking the polynomial $p(n) = n^2$, we conclude that μ is non-negligible.

- (c) (4 points) $\mu(n) := \mu'(n) \cdot p(n)$, for some negligible function $\mu'(n)$ and some polynomial function $p(n)$. Is such a $\mu(n)$ always negligible?

Solution

Negligible. Suppose for contradiction that $\mu(n) \geq 1/p'(n)$ for some polynomial $p'(n)$ for infinitely many n . Then, $\mu'(n) = \mu(n)/p(n) \geq 1/(p(n) \cdot p'(n))$ for infinitely

many n . Since $p(n) \cdot p'(n)$ is a polynomial, this shows that μ' is not negligible, which is a contradiction.

Problem 2. PRG or not? (12 points) Let $G : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{3n}$ be a pseudorandom generator (PRG). For each part below, either prove or disprove that the specified function G' is *necessarily* a PRG no matter which PRG G is used.

(a) (3 points)

$$G'(x) := G(\pi(x))$$

where $\pi : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{2n}$ is any $\text{poly}(n)$ -time computable bijective function. (You may not assume that π^{-1} is $\text{poly}(n)$ -time computable.)

Solution

PRG. To see why, suppose not for contradiction. Then, by negating the distinguisher if necessary, there is some PPT distinguisher D such that

$$\Pr_{x \leftarrow \{0,1\}^{2n}} [D(G'(x)) = 1] - \Pr_{y \leftarrow \{0,1\}^{3n}} [D(y) = 1] \geq 1/p(n)$$

for some polynomial $p(n)$ for infinitely many n . Since $G'(x) = G(\pi(x))$ and π is bijective, we have

$$\Pr_{x \leftarrow \{0,1\}^{2n}} [D(G'(x)) = 1] = \Pr_{x \leftarrow \{0,1\}^{2n}} [D(G(\pi(x))) = 1] = \Pr_{x' \leftarrow \{0,1\}^{2n}} [D(G(x')) = 1].$$

By relabeling x' back to x , we have

$$\Pr_{x \leftarrow \{0,1\}^{2n}} [D(G(x)) = 1] - \Pr_{y \leftarrow \{0,1\}^{3n}} [D(y) = 1] \geq 1/p(n)$$

for infinitely many n , which shows that G is not a PRG. This is a contradiction.

(b) (3 points)

$$G'(x||y) := G(x||x \oplus y)$$

where $x, y \in \{0, 1\}^n$. (**Note:** Here, and throughout the course, $x||y$ refers to the concatenation of two strings x and y .)

Solution

PRG. Define $\pi(x||y) = x||x \oplus y$. Then, $G'(z) = G(\pi(z))$, where $|z| = 2n$. If π is a $\text{poly}(n)$ -time computable bijection, then part (a) shows that G' is a PRG.

Clearly π is $\text{poly}(n)$ -time computable. Moreover, $\pi(\pi(x||y)) = \pi(x||x \oplus y) = x||x \oplus (x \oplus y) = x||y$, so π is its own inverse. Therefore, π is bijective.

(c) (3 points)

$$G'(s) := G(s)_{1,\dots,2n+1}$$

That is, the first $2n + 1$ bits of $G(s)$, so $G' : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{2n+1}$.

Solution

PRG. Suppose towards contradiction that G' is not a PRG. Then there exists a PPT distinguisher D such that

$$\left| \Pr_{y \leftarrow \{0,1\}^{2n+1}} [D(y) = 1] - \Pr_{s \leftarrow \{0,1\}^{2n}} [D(G'(s)) = 1] \right|$$

is non-negligible.

Construct a PPT distinguisher D' for G as follows: on input $z \in \{0, 1\}^{3n}$, run D on the first $2n + 1$ bits of z .

If z is uniform in $\{0, 1\}^{3n}$, then its first $2n + 1$ bits are uniform in $\{0, 1\}^{2n+1}$. If $z = G(s)$ for uniform $s \leftarrow \{0, 1\}^{2n}$, then its first $2n + 1$ bits are exactly $G'(s)$. Thus D' distinguishes $G(U_{2n})$ (where U_{2n} denotes the uniform distribution over strings of length $2n$) from U_{3n} with non-negligible advantage, contradicting the pseudorandomness of G .

(d) (3 points) Let $F : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{3n}$ be another PRG, and define

$$G'(s) := G(s) || F(s) .$$

Solution

Not a PRG. Consider the case $F = G$. Then

$$G'(s) = G(s) || G(s),$$

so the first and second halves of every output of G' are identical. Hence $G'(U_{2n})$ can be efficiently distinguished from the uniform distribution over $\{0, 1\}^{6n}$, and therefore G' is not necessarily a PRG.

Problem 3. Mixing PRGs. (6 points)

The cryptography community is debating between two different PRG candidates, G_1 and G_2 . Everyone agrees that at least one of them is secure, but they disagree on which it is. Can you make everyone happy by constructing a PRG out of G_1 and G_2 that is guaranteed to be secure assuming only that *at least* one of G_1 or G_2 is a PRG?

Explicitly, you may assume that the candidates G_1 and G_2 are polynomial-time computable functions expanding by one bit, and your goal is to come up with a PRG from G_1 and G_2 that has any non-trivial stretch (even one bit is fine).

Solution

First, we use the length-expansion construction from class to convert G_1 into $\tilde{G}_1$ that is length-tripling, using just G_1 . Similarly, we will convert G_2 into $\tilde{G}_2$ that is length-tripling, using just G_2 . This will make $\tilde{G}_1$ a length-tripling PRG if G_1 is a PRG, and this will make $\tilde{G}_2$ a length-tripling PRG if G_2 is a PRG. In any case, $\tilde{G}_1$ and $\tilde{G}_2$ are $\text{poly}(n)$ -time computable, length-tripling functions.

Now, we construct

$$G(x||y) := \tilde{G}_1(x) \oplus \tilde{G}_2(y),$$

where $|x| = |y| = n$ and $G : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{3n}$.

Suppose that G_1 is a PRG. Then, $\tilde{G}_1$ is a length-tripling PRG. Then, we show that G is a PRG. Suppose not for contradiction. Then, possibly by negating, we have some PPT distinguisher D such that

$$\Pr_{x||y \leftarrow \{0,1\}^{2n}} [D(G(x||y)) = 1] - \Pr_{z \leftarrow \{0,1\}^{3n}} [D(z) = 1] \geq 1/p(n)$$

for some polynomial $p(n)$ for infinitely many n . Now, consider the distinguisher $D'(z)$ defined by sampling $y \leftarrow \{0, 1\}^n$ and outputting $D(z \oplus \tilde{G}_2(y))$. Then, we have

$$\begin{aligned} & \Pr_{x \leftarrow \{0,1\}^n} [D'(\tilde{G}_1(x)) = 1] - \Pr_{z \leftarrow \{0,1\}^{3n}} [D'(z) = 1] \\ &= \Pr_{x \leftarrow \{0,1\}^n, y \leftarrow \{0,1\}^n} [D(\tilde{G}_1(x) \oplus \tilde{G}_2(y)) = 1] - \Pr_{z \leftarrow \{0,1\}^{3n}, y \leftarrow \{0,1\}^n} [D(z \oplus \tilde{G}_2(y)) = 1] \\ &= \Pr_{x||y \leftarrow \{0,1\}^{2n}} [D(G(x||y)) = 1] - \Pr_{z \leftarrow \{0,1\}^{3n}} [D(z) = 1] \geq 1/p(n), \end{aligned}$$

for infinitely many n . Therefore, $\tilde{G}_1$ is not a PRG, reaching our contradiction.

Similarly, if G_2 is a PRG, then G is a PRG by an analogous argument.

Problem 4. Hiding Promises. (14 points) We define a cryptographic protocol called a *promise hiding* scheme between two parties, Alice and Bob. Alice wants to promise something to Bob (for now, let's say one bit $\sigma \in \{0, 1\}$), but Alice wants this promise to be hidden from Bob until Alice decides the time is right to reveal σ .

- (a) (2 points) Suppose Alice samples $r \leftarrow \{0, 1\}$ and sends the message $A(\sigma; r) := \sigma \oplus r$ to Bob to promise $\sigma \in \{0, 1\}$. Later, when Alice decides the time is right, to reveal σ , Alice sends (σ, r) to Bob, and Bob can compute $A(\sigma; r)$ to verify that it is consistent with the message he received from Alice. We would like two properties to hold:

- **(Statistically) Hiding:** The distributions $A(0; r)$ and $A(1; r)$ are identical for random r .
- **(Statistically) Promising:** There does not exist $r_0, r_1 \in \{0, 1\}$ such that $A(0; r_0) = A(1; r_1)$.

Prove or disprove that this scheme is hiding, and prove or disprove that this scheme is promising.

Solution

This scheme is hiding but not promising.

To see it is hiding, $A(\sigma; r)$ is a one-time pad of σ with key r .

To see that it is not promising, let $r_0 = 0$ and $r_1 = 1$. Then, $0 \oplus r_0 = 0 = 1 \oplus r_1$.

- (b) (4 points) Let G be any length-tripling PRG. Suppose Alice samples $r \leftarrow \{0, 1\}^n$ and sends the message

$$A(\sigma; r) = \begin{cases} G(r) & \text{if } \sigma = 0, \\ G(r) \oplus 1^{3n} & \text{if } \sigma = 1. \end{cases}$$

We would like similar properties to hold:

- **(Computationally) Hiding:** The distributions $A(0; r)$ and $A(1; r)$ are computationally indistinguishable for random r .
- **(Statistically) Promising:** There does not exist $r_0, r_1 \in \{0, 1\}^n$ such that $A(0; r_0) = A(1; r_1)$.

Prove or disprove that this scheme is hiding, and prove or disprove that this scheme is necessarily promising.

Solution

This scheme is hiding but not promising.

To see it is hiding, first observe that if $\sigma = 0$, then $G(r)$ looks indistinguishable from uniformly random $\{0, 1\}^{3n}$, by the definition of a PRG. Similarly, if $G(r) \oplus 1^{3n}$ looks indistinguishable from uniformly random $\{0, 1\}^{3n}$, then by the triangle inequality, computational hiding holds.

We now show that $G(r) \oplus 1^{3n}$ looks indistinguishable from uniformly random $\{0, 1\}^{3n}$. Suppose for contradiction that we have a PPT distinguisher D such that, possibly by negating D ,

$$\Pr_{r \leftarrow \{0, 1\}^n} [D(G(r) \oplus 1^{3n}) = 1] - \Pr_{z \leftarrow \{0, 1\}^{3n}} [D(z) = 1] \geq 1/p(n)$$

for some polynomial $p(n)$ and infinitely many n . Then, define the distinguisher $D'(z) := D(z \oplus 1^{3n})$. We have

$$\begin{aligned} & \Pr_{r \leftarrow \{0, 1\}^n} [D'(G(r)) = 1] - \Pr_{z \leftarrow \{0, 1\}^{3n}} [D'(z) = 1] \\ &= \Pr_{r \leftarrow \{0, 1\}^n} [D(G(r) \oplus 1^{3n}) = 1] - \Pr_{z \leftarrow \{0, 1\}^{3n}} [D(z \oplus 1^{3n}) = 1] \\ &= \Pr_{r \leftarrow \{0, 1\}^n} [D(G(r) \oplus 1^{3n}) = 1] - \Pr_{z \leftarrow \{0, 1\}^{3n}} [D(z) = 1] \geq 1/p(n), \end{aligned}$$

for infinitely many n . Therefore, we have a distinguisher for G , which is a contradiction.

To see that it is not promising, let H be a length-tripling PRG. Then, let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{3n}$ be defined by

$$G(r) := \begin{cases} 0^{3n} & \text{if } r = 0^n, \\ 1^{3n} & \text{if } r = 1^n, \\ H(r) & \text{otherwise.} \end{cases}$$

Since G differs from H on only the two inputs 0^n and 1^n , the statistical distance between $G(U_n)$ and $H(U_n)$ is at most $2/2^n$, which is negligible. Hence G is a PRG whenever H is. However, for this G , let $r_0 = 0^n$ and $r_1 = 1^n$, and note that

$$A(0; r_0) = G(0^n) = 0^{3n} = 1^{3n} \oplus 1^{3n} = G(1^n) \oplus 1^{3n} = A(1; r_1),$$

so this scheme is not promising.

(c) (2 points) Let G be any length-tripling function. Show that

$$\Pr_{b \leftarrow \{0,1\}^{3n}} [\exists r_0, r_1 \in \{0,1\}^n : b = G(r_0) \oplus G(r_1)] \leq \frac{1}{2^n}.$$

Solution

Let $S = \{G(r_0) \oplus G(r_1) : r_0, r_1 \in \{0,1\}^n\}$. Clearly $|S| \leq 2^n \cdot 2^n = 2^{2n}$. Therefore,

$$\Pr_{b \leftarrow \{0,1\}^{3n}} [b \in S] = \frac{|S|}{2^{3n}} \leq \frac{1}{2^n}.$$

The claim follows.

(d) (6 points) Let's change things up a bit. Assume the existence of a length-tripling PRG G . Now, suppose that Bob is allowed to send a uniformly random string $b \leftarrow \{0,1\}^{3n}$ before Alice sends her message. Then, Alice's message can be described as a function $A(b, \sigma; r)$, so that it can depend on b as well. Consider the following promise hiding scheme for Alice and Bob:

$$A(b, \sigma; r) = \begin{cases} G(r) & \text{if } \sigma = 0, \\ G(r) \oplus b & \text{if } \sigma = 1. \end{cases}$$

Prove that it satisfies the two properties below:

- **(Computationally) Hiding:**¹ For all $b \in \{0,1\}^{3n}$, the distributions $A(b, 0; r)$ and $A(b, 1; r)$ are computationally indistinguishable for random r .
Prove this via a hybrid argument and make sure to clearly define your hybrids and prove formally that any two adjacent hybrids are computationally indistinguishable.
- **(Statistically) Promising:**

$$\Pr_{b \leftarrow \{0,1\}^{3n}} [\exists r_0, r_1 \in \{0,1\}^n : A(b, 0; r_0) = A(b, 1; r_1)] \leq \text{negl}(n).$$

¹For this part, we allow the distinguishers to be non-uniform: they may have a polynomial-size advice string depending only on the input length. In particular, a fixed string $b = b_n$ may be hardwired into the (PRG) distinguisher.

Solution

We first show computational hiding. Fix any $b \in \{0, 1\}^{3n}$ and consider the following hybrids:

$$H_0 = G(U_n), \quad H_1 = U_{3n}, \quad H_2 = U_{3n} \oplus b, \quad H_3 = G(U_n) \oplus b.$$

By the pseudorandomness of G , we have

$$H_0 \approx_c H_1,$$

where $\approx_c$ denotes computational indistinguishability. Moreover, H_1 and H_2 are identically distributed, since XORing a uniformly random string with a fixed string preserves uniformity. Finally,

$$H_2 \approx_c H_3$$

as well: indeed, if a non-uniform polynomial-time distinguisher D (with b hardwired) distinguishes H_2 from H_3 , then the distinguisher $D'(z) = D(z \oplus b)$ distinguishes U_{3n} from $G(U_n)$ with the same advantage, contradicting the pseudorandomness of G . Therefore, by the hybrid argument,

$$A(b, 0; U_n) = H_0 \approx_c H_3 = A(b, 1; U_n).$$

Promising holds directly due to the previous part (by XORing by $G(r_1)$ on both sides of the expression in (c)).